



Vessel Speed Optimization Under Oil Price Volatility

Strategic analysis of speed decisions in container shipping when fuel prices face geopolitical risk

Motivation and Background

Why This Matters

Vessel speed represents one of the most critical operational tradeoffs in container shipping economics. The decision directly impacts both operational costs and service quality.

- Faster sailing reduces transit time but sharply increases fuel consumption
- Slower sailing lowers bunker costs but increases inventory-in-transit carrying costs
- On fixed weekly services, excessive slow steaming may require deploying an additional vessel

Why Now

Current market conditions create unprecedented fuel price volatility that demands strategic reassessment of speed policies.

- Oil prices are highly exposed to geopolitical disruption and supply chain shocks
- The Strait of Hormuz is a major global oil chokepoint



Data source: U.S. Energy Information Administration, *Short-Term Energy Outlook* (STEO), April 2026

Operational Insight: The lowest-fuel-speed is not always the lowest-total-cost speed. Optimal speed must balance fuel burn, transit time, and fleet deployment constraints.

Research Question

What is the cost-minimizing vessel speed as oil prices change, and how does the fleet-frequency constraint shift that optimum?

Layer 1 Base Speed Optimization

Decision variable: v = vessel speed (knots), subject to $12 \leq v \leq 25$

Parameters

D	<i>One – way route distance (Shanghai to LA)</i>
$v_0 = 24 \text{ kn}$	<i>Design (reference) speed</i>
p	<i>Bunker fuel price (\$/ MT)</i>
α	<i>Fuel consumption at v_0 (MT/ day)</i>
h	<i>Daily pipeline inventory holding cost (\$/ day)</i>

Optimization Model

Layer 1 Base Speed Optimization

Decision variable: v = vessel speed (knots), subject to $12 \leq v \leq 25$

Fuel consumption per day (cubic admiralty law):

$$FC(v) = \alpha \cdot \left(\frac{v}{v_0}\right)^3 [MT/day]$$

Transit time (one way; 24 converts knots to nmi/ day):

$$T(v) = \frac{D}{24v} [days]$$

Cost components per voyage:

$$C_{fuel}(v) = p \cdot FC(v) \cdot T(v) = \frac{p\alpha D}{24v_0^3} v^2$$

$$C_{inv}(v) = h \cdot T(v) = \frac{hD}{24v}$$

Optimization Model

Total cost per voyage:

$$TC(v) = \frac{p\alpha D}{24v_0^3} v^2 + \frac{hD}{24} v^{-1} = A v^2 + B v^{-1}$$

Closed-form optimal speed (set $\frac{dTC}{dv} = 0$):

$$v^* = \left[\frac{h v_0^3}{2p\alpha} \right]^{1/3}$$

Key result: $v^* \propto p^{-1/3}$. If bunker price doubles, optimal speed falls by $2^{1/3} \sim 21\%$.

Optimization Model

Layer 2 Fleet frequency constraint

To maintain weekly departures with ~ 2 days port time each end, the number of vessels required is:

$$T_{round}(v) = \frac{2D}{24v} + 4 \quad [days]$$

$$N(v) = \left\lceil \frac{T_{round}(v)}{7} \right\rceil \quad [vessels]$$

Each vessel carries weekly charter cost c_{cap} (\$/vessel/week). c_{op} is the operating cost to maintain the fleet including crew.

Extended weekly total cost:

$$TC_{ext}(v) = \underbrace{2(Av^2 + Bv^{-1})}_{\text{One-way transit cost}} + N(v) \cdot (c_{cap} + 7c_{op})$$

Key Parameter Values

<i>Parameter</i>	<i>Symbol</i>	<i>Value</i>
<i>Route distance</i>	D	<i>5,500 nmi</i>
<i>Design speed</i>	v_0	<i>24 kn</i>
<i>Fuel at design speed</i>	α	<i>165 MT/day</i>
<i>Bunker price</i>	p	<i>\$1000–\$3,000/MT</i>
<i>Holding cost rate</i>	i	<i>20%/yr</i>
<i>Cargo value per TEU</i>	—	<i>\$30,000</i>
<i>Loaded TEUs</i>	—	<i>10,200</i>
<i>Daily holding cost</i>	h	<i>\$168,000/day</i>
<i>Daily OPEX</i>	c_{op}	<i>\$20,000/day</i>
<i>Charter cost</i>	c_{cap}	<i>\$455,000/week</i>
<i>Port time</i>	—	<i>2 days/port</i>

Simulation and Inference

Bunker Price (\$/MT)	L1 Opt Speed (kn)	L2 Opt Speed (kn)	Fleet Size (N)	Weekly Cost (\$M)	Transit Time (Days)
1000	19.16	19.16	4	8.407	12.0
1500	16.74	19.11	4	9.406	12.0
2000	15.21	19.11	4	10.404	12.0
2500	14.12	14.80	5	11.173	15.5
3000	13.29	14.80	5	11.772	15.5

Slowing down ships to save fuel only works until it breaks the weekly schedule. Once a trip takes too long, operators are forced to charter an extra ship just to keep port calls on time.

⚠ That extra ship costs nearly \$600,000 a week, completely wiping out any fuel savings. Because of this massive penalty, practical speeds are locked by the schedule, not by fuel prices.

Fuel would have to reach an extreme \$2,220 per ton to make a slower, larger fleet profitable. Until then, shipping lines will always prioritize their schedules and capital efficiency over fuel conservation.

Fig 1. Weekly System Cost (The Sawtooth Effect)

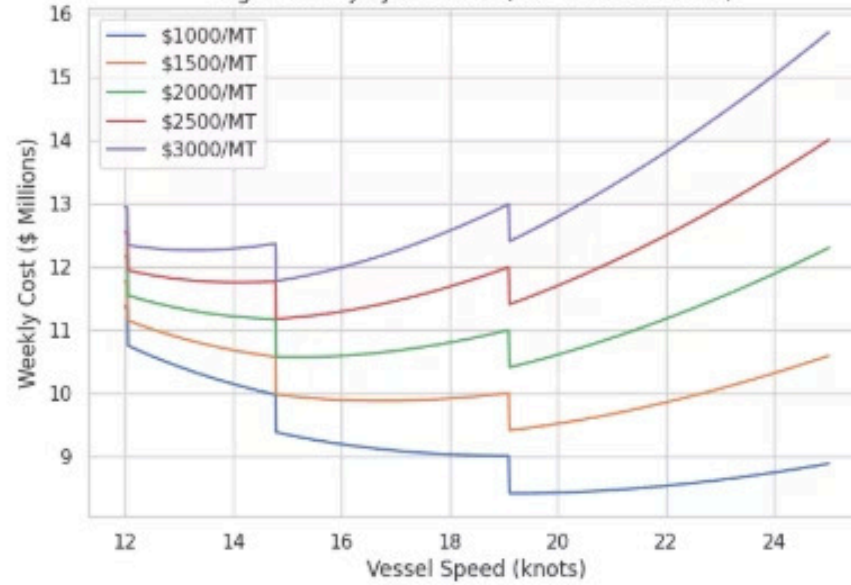


Fig 2. Fleet Requirements vs. Speed

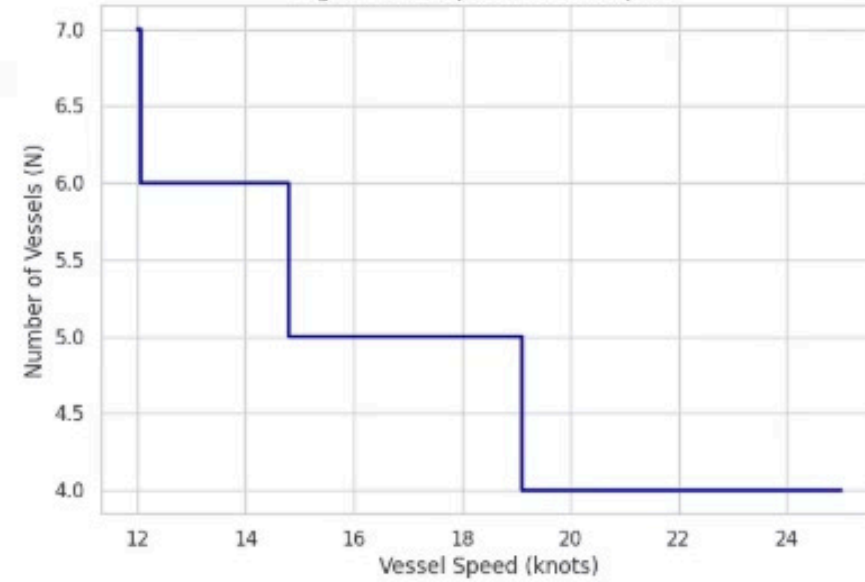


Fig 3. Optimal Speed Response (Operational Rigidity)

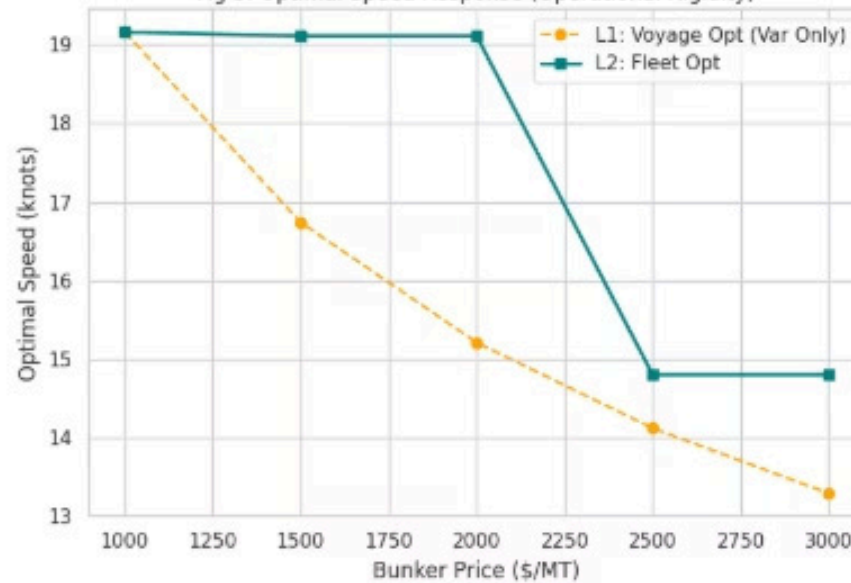
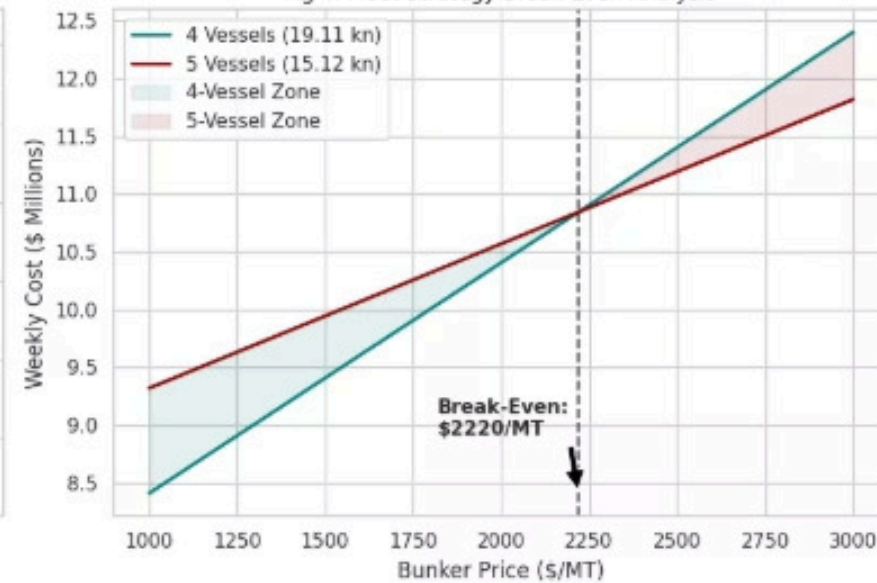


Fig 4. Fleet Strategy Break-Even Analysis



Connection to Course Concepts

1

EOQ Trade-Off

- The model follows the same core logic as EOQ: balancing opposing costs to find an optimal decision.
- Here, fuel cost increases with vessel speed, while inventory holding cost decreases as faster travel reduces transit time.
- The optimal speed comes from the same first-order condition logic.

2

Pipeline Inventory Interpretation

- Goods in transit are treated as pipeline inventory.
- Sailing time acts as the lead time.
- The holding cost term captures the cost of inventory tied up at sea, directly linking the model to course concepts on lead times and in-transit stock.

3

Fixed-Cost Fleet Structure

- In Layer 2, adding another vessel creates a discrete fixed cost.
- This produces a step-cost structure and makes the optimization non-smooth.
- The setup is similar to fixed-charge network design / facility location, where adding capacity introduces non-convexities.

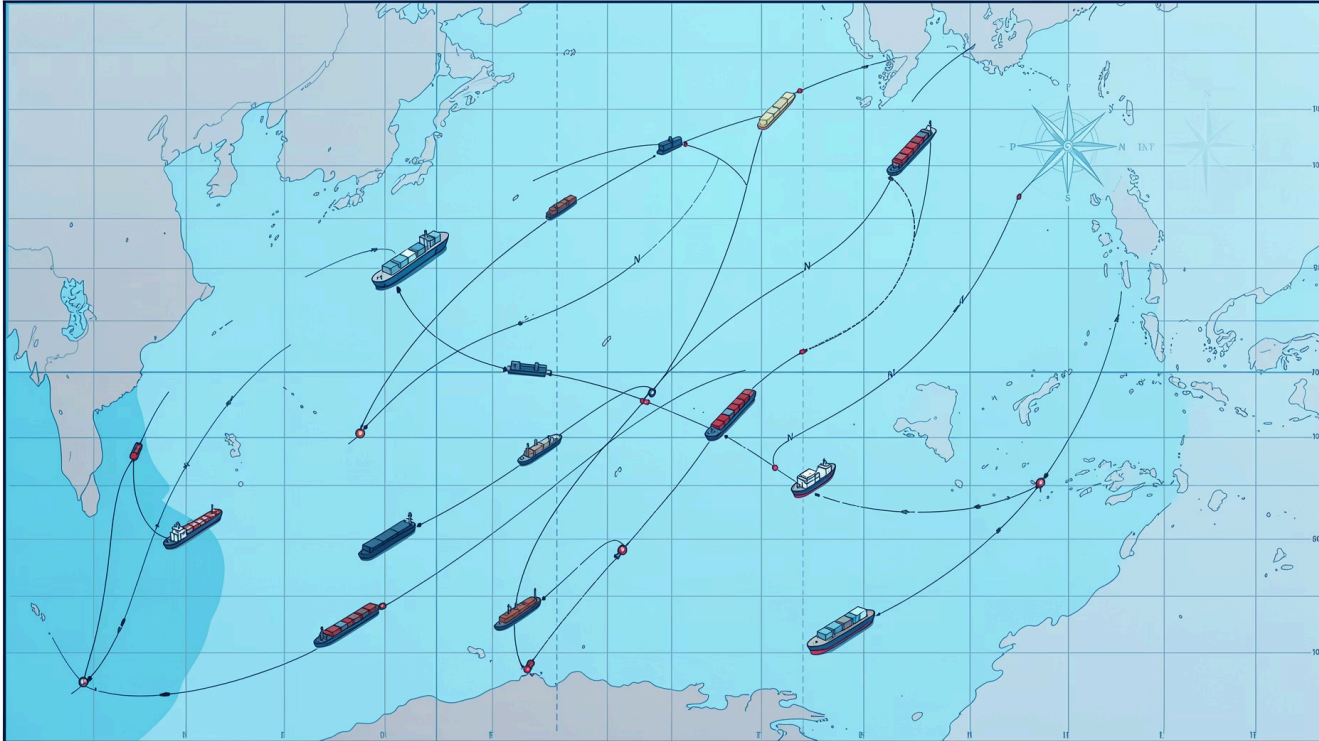
Appendix

Route Distance

D = 5,500 nautical miles

Fixed navigational distance along the **North Pacific Great Circle lane**, sourced from SeaRates.com. This is the actual nautical distance from Shanghai (Yangshan Port) to Los Angeles (Port of Long Beach) along the North Pacific shipping lane.

 Source: SeaRates.com | Type: Fixed constant



PARAMETER 2 OF 11

Design Speed

$v_o = 24$ knots

The vessel's rated service speed per This is the speed the vessel was engineered to sail at maximum engine load. It is taken directly from Maersk's published specifications for large containerships on this route.

PARAMETER 3 OF 11

Fuel at Design Speed

$\alpha = 165$ MT/day

Main engine fuel consumption at full design speed (24 kn). Set at the **midpoint of the Wärtsilä-published range of 150–180 MT/day** for ultra-large container vessels. We take the midpoint (165) as a neutral base case. If we used the lower bound we would underestimate fuel costs, the upper bound would overestimate. Sensitivity analysis tests both ends.

 Source: Wärtsilä engine specs | Type: Base-case constant

PARAMETER 5 OF 11

Holding Cost Rate

i = 20% per year

The annualized cost of holding cargo in transit, set at the **lower bound of the 20–25% range** published in the *Journal of Business Logistics*. Includes capital tied up, insurance and risk. The conservative 20% assumption prevents overstating the time-cost penalty of slow steaming.

 Source: Journal of Business Logistics | Conservative assumption

PARAMETER 6 OF 11

Cargo Value per TEU

\$30,000 / TEU

US Census Bureau trade data shows the average value of containerized goods shipped from Asia to the US. The \$30,000 per TEU figure reflects a realistic mix of electronics, apparel, and industrial goods (not a premium or budget assumption). It feeds directly into the holding cost calculation.

 Source: US Census Bureau | Mixed Asia-US manifest

Loaded TEUs & Daily Holding Cost

PARAMETER 7 OF 11

Loaded TEUs = 10,200

The vessel has a capacity of 12,000 TEU but ships rarely sail full. An 85% load factor is the industry standard for trans-Pacific services. So $12,000 \times 0.85 = 10,200$ TEUs actually loaded. Using 100% would overstate how much cargo is at risk; using a lower figure would understate it.

PARAMETER 8 OF 11

Daily Holding Cost $h = \$168,000/\text{day}$

This is not sourced directly — it is calculated from the three parameters:

$$h = i \times \text{Value} \times \text{TEUs} / 365$$

$$= 0.20 \times \$30,000 \times 10,200 / 365 \approx \$168,000/\text{day}$$

Operating & Charter Costs

Daily OPEX — $c_{op} = \$20,000/\text{day}$

Drewry's annual benchmarking report covers crew, maintenance, and insurance costs for large container ships. The range is \$15,000–\$25,000/day. We take the midpoint. Like holding cost, this is a time-based cost, every extra day at sea adds \$20,000 in running costs regardless of speed.

Charter Cost — $c_{cap} = \$65,000/\text{day}$

This only activates in Layer 2 when slow steaming forces us to add an extra ship to maintain weekly departures. Clarksons reports \$50,000–\$80,000/day for this vessel class in the current market. We use the midpoint (\$65,000/day), converted to weekly since the model counts vessels per week: $\$65,000 \times 7 = \$455,000$.

Port Time

2 days per port call

AIS tracking data from MarineTraffic shows typical port turnaround times for large containerships at Shanghai and Long Beach are approximately 2 days each, covering berthing, loading, and unloading. With two port calls per round trip, this adds a fixed 4 days to every loop regardless of how fast the ship sails.

 Source: MarineTraffic AIS data | Fixed regardless of speed

